Questions
Lecture – 5

1. What is the electromagnetic induction?
2. State and explain Faradays law of inductions. Write down the mathematical form of Newman's law.
3. State the Lenz's law.
4. Explain the energy conservation in electromagnetic induction.
5. Write a short note on: i) Self-induction ii) Mutual induction
6. Define 1 henry.
7. Explain the basic construction transformer.
8. Magnetic flux of $3 \times 10^{-8} wb$ crosses through a coil having 500 turns. The flux decreases to $3 \times 10^{-3} wb$ in 0.015. What is the induced $e.m.f.$?
9. When the current of $2A$ flows through a coil of 400 turns creates magnetic flux of $4 \times 10^{-4} wb$. Calculate the self – inductance of the coil.
10. Magnetic flux in a coil of 400 turns changes from $8 \times 10^{-5} wb$ to $50 \times 10^{-5} wb$ in $3 \times 10^{-3} sec$. Calculate the induced emf in the coil.
11. Average emf induced in a coil is $32V$ when current decreases from $10A$ to $2A$ in $0.1 sec$ in the coil. Calculate the self – inductance in the coil.
12. If the current in the primary coil is changed from $6A$ to $1A$ in $0.05 sec$, then emf $5V$ is induced in secondary coil. What is the mutual inductance of the coil?
13. Two nearby coil A and B have number of turns respectively of 200 and 1000. When $2A$ current flows through the coil, then magnetic flux $2.4 \times 10^{-4} wb$ is produced in the coil A and $1.6 \times 10^{-4} wb$ in the coil B. Calculate (i) the self – inductance of the coil A (ii) the mutual inductance of the coil B. (iii) If current stops A in 0.4 sec, find the inducted emf in coil B.